

Formalization of the Multisegment Construction

Let u, v, w denote mathematical objects.

Definition 1.

$$\emptyset = \{u \mid u \neq u\}.$$

Let the empty set denote $\emptyset$.

Axiom 2 (Set Existence). $\emptyset$ is a set.

Definition 3. Let C be a class.

A subset of C is a set X such that every element of X is an element of C .

Let $X \subseteq C$ stand for X is a subset of C .

Axiom 4 (Separation). Let X be a set. Let Y be a class such that every element of Y is an element of X .

Then Y is a set.

Signature 5. An integer is a mathematical object.

Let a, b, c, n denote integers.

Definition 6. $\mathbb{Z}$ is the collection of integers.

Axiom 7. $\mathbb{Z}$ is a set.

Signature 8. A natural number is an integer.

Let k, r, s denote natural numbers.

Definition 9. $\mathbb{N}$ is the collection of natural numbers.

Axiom 10. $\mathbb{N}$ is a set.

Signature 11. 0 is a natural number.

Signature 12. 1 is a natural number.

Signature 13. $a + b$ is an integer.

Signature 14. $a - b$ is an integer.

Axiom 15. Let r, s be natural numbers.

$r + s$ is a natural number.

Definition 16.

$$a \leq b \iff \text{there exists a natural number } k \text{ such that } a + k = b.$$

Let $a < b$ stand for $a \leq b$ and $a \neq b$.

Axiom 17. If $a \leq b \leq c$, then $a \leq c$.

Definition 18. $[a, b]$ is the collection of integers n such that

$$a \leq n \leq b.$$

Lemma 19. $[a, b]$ is a set.

Axiom 20.

$$[a + 1, a] = \emptyset.$$

Definition 21. A segment is a set Δ such that there exists an integer a and an integer b such that

$$a \leq b$$

and

$$\Delta = [a, b].$$

Definition 22. Let Δ be a segment.

$b(\Delta)$ is an integer a such that there exists an integer b such that

$$a \leq b$$

and

$$\Delta = [a, b].$$

Definition 23. Let Δ be a segment.

$e(\Delta)$ is an integer b such that there exists an integer a such that

$$a \leq b$$

and

$$\Delta = [a, b].$$

Definition 24. Let Δ be a segment.

$$\overleftarrow{\Delta} = [b(\Delta) - 1, e(\Delta) - 1].$$

Definition 25. Let Δ be a segment.

$$\overrightarrow{\Delta} = [b(\Delta) + 1, e(\Delta)].$$

Lemma 26. Let Δ be a segment.

If

$$b(\Delta) = e(\Delta),$$

then

$$\overrightarrow{\Delta} = \emptyset.$$

Definition 27. Seg is the collection of segments.

Definition 28. Let Δ and Γ be segments.

$$\Delta \prec \Gamma$$

if and only if

$$b(\Delta) < b(\Gamma)$$

and

$$e(\Delta) < e(\Gamma).$$

Signature 29. A multisegment is a mathematical object.

Let m denote a multisegment.

Signature 30. Let m be a multisegment.

$\text{size}(m)$ is a natural number.

Let i, j denote natural numbers.

Definition 31. Let m be a multisegment.

$I(m)$ is the collection of natural numbers i such that

$$1 \leq i \leq \text{size}(m).$$

Lemma 32. Let m be a multisegment.

$I(m)$ is a set.

Signature 33. Let m be a multisegment. Let i be an element of $I(m)$.

Δ_i^m is a segment.

Let C denote a mathematical object.

Proposition 34. Assume that C is a map and

$$u \in \text{dom}(C).$$

Then $C(u)$ is a mathematical object.

Axiom 35. Assume that C is a function.

Then $\text{dom}(C)$ is a set.

Let the domain of C denote $\text{dom}(C)$.

Definition 36. Let m be a multisegment. Let C be a function.

$V(m, C)$ is the collection of mathematical objects u such that

$$u \in \text{dom}(C)$$

and

$$C(u) \in I(m).$$

Definition 37. Let C be a function.

$S(C)$ is the collection of natural numbers r such that

$$r \in \text{dom}(C)$$

and

$$r + 1 \in \text{dom}(C).$$

Lemma 38. Let C be a function.

$S(C)$ is a set.

Definition 39. Let m be a multisegment. Let C be a function.

$G(m, C)$ is the collection of natural numbers r such that

$$r \in S(C),$$

$$C(r) \in I(m),$$

$$C(r+1) \in I(m),$$

$$b(\Delta_{C(r)}^m) < b(\Delta_{C(r+1)}^m),$$

and

$$e(\Delta_{C(r)}^m) < e(\Delta_{C(r+1)}^m).$$

Definition 40. Let m be a multisegment. Let i be an element of $I(m)$.

$J(m, i)$ is the collection of natural numbers j such that there exists a function C satisfying

$$\text{dom}(C) = [0, j],$$

$$0 \in \text{dom}(C),$$

$$C(0) = i,$$

$$\text{dom}(C) \subseteq V(m, C),$$

and

$$S(C) \subseteq G(m, C).$$

Definition 41. Let k be a natural number.

$B(k)$ is the collection of natural numbers r such that

$$r \leq k.$$

Lemma 42. Let k be a natural number.

$B(k)$ is a set.

Definition 43. Let m be a multisegment. Let i be an element of $I(m)$.

$d_m(i)$ is a natural number k such that

$$k \in J(m, i)$$

and

$$J(m, i) \subseteq B(k).$$

Definition 44. Let m be a multisegment.

$D(m)$ is the collection of natural numbers k such that there exists an element i of $I(m)$ with

$$k = d_m(i).$$

Lemma 45. Let m be a multisegment.

$D(m)$ is a set.

Definition 46. Let m be a multisegment such that

$$D(m) \neq \emptyset.$$

$d(m)$ is a natural number k such that

$$k \in D(m)$$

and

$$D(m) \subseteq B(k).$$

Definition 47. Let m be a multisegment. Let k be a natural number.

I_k^m is the collection of natural numbers i such that

$$i \in I(m)$$

and

$$d_m(i) = k.$$

Lemma 48. Let m be a multisegment. Let k be a natural number.

I_k^m is a set.

Axiom 49 (Same Depth Nesting). Let m be a multisegment. Let k be a natural number. Let i, j be elements of I_k^m .

Then

$$\Delta_i^m \subseteq \Delta_j^m$$

or

$$\Delta_j^m \subseteq \Delta_i^m.$$

Signature 50. Let m be a multisegment. Let k be a natural number such that

$$I_k^m \neq \emptyset.$$

$\ell_m(k)$ is a natural number.

Axiom 51 (Positive Enumeration Length). Let m be a multisegment. Let k be a natural number such that

$$I_k^m \neq \emptyset.$$

Then

$$1 \leq \ell_m(k).$$

Signature 52. Let m be a multisegment. Let k be a natural number such that

$$I_k^m \neq \emptyset.$$

Let r be a natural number.

$\iota_{m,k}(r)$ is a natural number.

Axiom 53 (Enumeration Entries). Let m be a multisegment. Let k be a natural number such that

$$I_k^m \neq \emptyset.$$

Let r be a natural number such that

$$1 \leq r \leq \ell_m(k).$$

Then

$$\iota_{m,k}(r) \in I_k^m.$$

Axiom 54 (Enumeration Coverage). Let m be a multisegment. Let k be a natural number such that

$$I_k^m \neq \emptyset.$$

Let i be an element of I_k^m .

There exists a natural number r such that

$$1 \leq r \leq \ell_m(k)$$

and

$$\iota_{m,k}(r) = i.$$

Axiom 55 (Enumeration Injectivity). Let m be a multisegment. Let k be a natural number such that

$$I_k^m \neq \emptyset.$$

Let r, s be natural numbers such that

$$1 \leq r \leq \ell_m(k)$$

and

$$1 \leq s \leq \ell_m(k).$$

If

$$\iota_{m,k}(r) = \iota_{m,k}(s),$$

then

$$r = s.$$

Axiom 56 (Admissible Enumeration). Let m be a multisegment. Let k be a natural number such that

$$I_k^m \neq \emptyset.$$

Let r be a natural number such that

$$1 \leq r < \ell_m(k).$$

Assume that

$$\iota_{m,k}(r) \in I_k^m.$$

Assume that

$$\iota_{m,k}(r+1) \in I_k^m.$$

Then

$$\Delta_{\iota_{m,k}(r+1)}^m \subseteq \Delta_{\iota_{m,k}(r)}^m.$$

Signature 57. Let m be a multisegment. Let k be a natural number such that

$$I_k^m \neq \emptyset.$$

j_k^m is a natural number.

Axiom 58 (Final Enumeration Index). Let m be a multisegment. Let k be a natural number such that

$$I_k^m \neq \emptyset.$$

Then

$$j_k^m = \iota_{m,k}(\ell_m(k)).$$

Axiom 59 (Final Index Membership). Let m be a multisegment. Let k be a natural number such that

$$I_k^m \neq \emptyset.$$

Then

$$j_k^m \in I_k^m.$$

Lemma 60. Let m be a multisegment. Let k be an element of $D(m)$.

Then

$$I_k^m \neq \emptyset.$$

Definition 61. Let m be a multisegment.

$I'(m)$ is the collection of natural numbers i such that

$$i \in I(m)$$

and there exists a natural number k such that

$$k \in D(m),$$

$$I_k^m \neq \emptyset,$$

and

$$i = j_k^m.$$

Lemma 62. Let m be a multisegment.

$I'(m)$ is a set.

Definition 63. Let m be a multisegment.

$I''(m)$ is the collection of natural numbers i such that

$$i \in I(m)$$

and

$$i \notin I'(m).$$

Lemma 64. Let m be a multisegment.

$I''(m)$ is a set.

Signature 65. Let m be a multisegment. Let i be an element of $I(m)$.

$i_m^\vee$ is a natural number.

Axiom 66 (Vee Preserves Depth Fibre). Let m be a multisegment. Let k be a natural number. Let i be an element of I_k^m .

Then

$$i_m^\vee \in I_k^m.$$

Axiom 67 (Vee Successor). Let m be a multisegment. Let k be a natural number such that

$$I_k^m \neq \emptyset.$$

Let r be a natural number such that

$$1 \leq r < \ell_m(k).$$

Assume that

$$\iota_{m,k}(r) \in I_k^m.$$

Assume that

$$\iota_{m,k}(r+1) \in I_k^m.$$

Then

$$\iota_{m,k}(r)_m^\vee = \iota_{m,k}(r+1).$$

Axiom 68 (Vee Wrap Around). Let m be a multisegment. Let k be a natural number such that

$$I_k^m \neq \emptyset.$$

Assume that

$$j_k^m \in I_k^m.$$

Assume that

$$\iota_{m,k}(1) \in I_k^m.$$

Then

$$j_k^{m\vee} = \iota_{m,k}(1).$$

Axiom 69 (Vee Injectivity). Let m be a multisegment. Let i, j be elements of $I(m)$.

If

$$i_m^\vee = j_m^\vee,$$

then

$$i = j.$$

Axiom 70 (Vee Surjectivity). Let m be a multisegment. Let j be an element of $I(m)$.

There exists an element i of $I(m)$ such that

$$i_m^\vee = j.$$

Signature 71. Let m be a multisegment. Let i be an element of $I(m)$.

Δ_i^m is a segment.

Axiom 72 (Transformed Segment). Let m be a multisegment. Let i be an element of $I(m)$.

$$\Delta_i^m = [\text{b}(\Delta_i^m), \text{e}(\Delta_{i_m^\vee}^m)].$$

Axiom 73 (Transformed Segment Beginning). Let m be a multisegment. Let i be an element of $I(m)$.

$$\text{b}(\Delta_i^m) = \text{b}(\Delta_i^m).$$

Axiom 74 (Transformed Segment End). Let m be a multisegment. Let i be an element of $I(m)$.

$$\text{e}(\Delta_i^m) = \text{e}(\Delta_{i_m^\vee}^m).$$

Axiom 75 (Depth Values Form an Interval). Let m be a multisegment such that

$$D(m) \neq \emptyset.$$

Then

$$D(m) = [0, d(m)].$$

Signature 76. Let m be a multisegment such that

$$D(m) \neq \emptyset.$$

$l(m)$ is a multisegment.

Axiom 77 (Ladder Size). Let m be a multisegment such that

$$D(m) \neq \emptyset.$$

Then

$$\text{size}(l(m)) = d(m) + 1.$$

Axiom 78 (Ladder Index). Let m be a multisegment such that

$$D(m) \neq \emptyset.$$

Let k be an element of $D(m)$.

Then

$$k + 1 \in I(l(m)).$$

Axiom 79 (Ladder Entry). Let m be a multisegment such that

$$D(m) \neq \emptyset.$$

Let k be an element of $D(m)$.

Assume that

$$k + 1 \in I(l(m)).$$

Assume that

$$I_k^m \neq \emptyset.$$

Then

$$\Delta_{k+1}^{l(m)} = \Delta_{j_k^m}^m.$$

Signature 80. Let m be a multisegment.

$q(m)$ is a natural number.

Axiom 81 (Empty Remaining Enumeration). Let m be a multisegment.

If

$$I''(m) = \emptyset,$$

then

$$q(m) = 0.$$

Axiom 82 (Positive Remaining Enumeration Length). Let m be a multisegment.

If

$$I''(m) \neq \emptyset,$$

then

$$1 \leq q(m).$$

Signature 83. Let m be a multisegment. Let r be a natural number.

$\rho_m(r)$ is a natural number.

Axiom 84 (Remaining Enumeration Entries). Let m be a multisegment. Let r be a natural number such that

$$1 \leq r \leq q(m).$$

Then

$$\rho_m(r) \in I''(m).$$

Axiom 85 (Remaining Enumeration Coverage). Let m be a multisegment. Let i be an element of $I''(m)$.

There exists a natural number r such that

$$1 \leq r \leq q(m)$$

and

$$\rho_m(r) = i.$$

Axiom 86 (Remaining Enumeration Injectivity). Let m be a multisegment. Let r, s be natural numbers such that

$$1 \leq r \leq q(m)$$

and

$$1 \leq s \leq q(m).$$

If

$$\rho_m(r) = \rho_m(s),$$

then

$$r = s.$$

Signature 87. Let m be a multisegment.

m' is a multisegment.

Axiom 88 (Derived Multisegment Size). Let m be a multisegment.

$$\text{size}(m') = q(m).$$

Axiom 89 (Derived Multisegment Entry). Let m be a multisegment. Let r be an element of $I(m')$.

Assume that

$$I''(m) \neq \emptyset.$$

Assume that

$$\rho_m(r) \in I''(m).$$

Then

$$\Delta_r^{m'} = \Delta_{\rho_m(r)}^m.$$